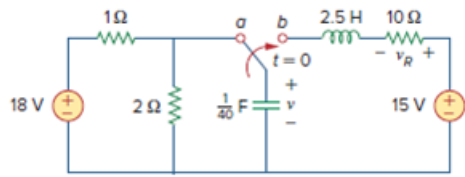


Problem

Having been in position *a* for a long time, the switch in Fig. 8.21 is moved to position *b* at $t = 0$. Find $v(t)$ and $v_R(t)$ for $t > 0$.

Figure 8.21:



Step-by-step solution

Step 1 of 10

Refer to Figure 8.21 from the textbook.

Draw the modified circuit for $t < 0$.

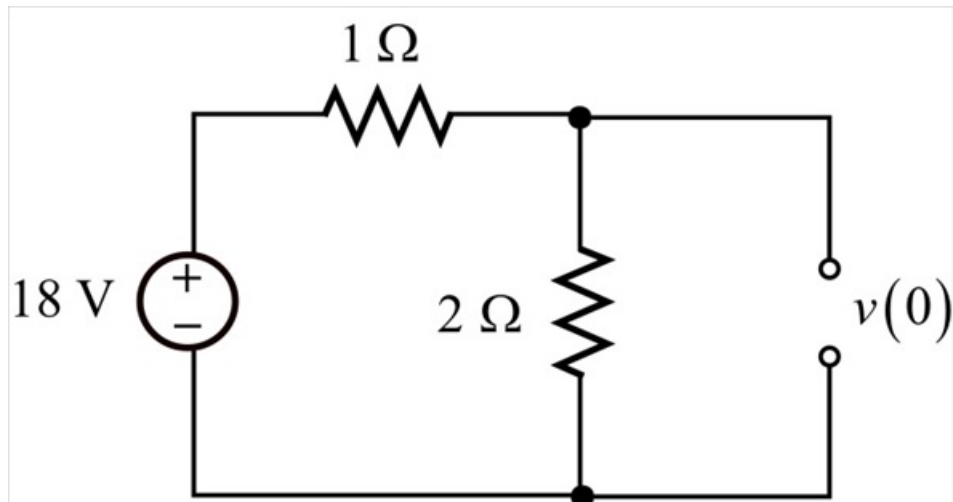


Figure 1

Step 2 of 10

Apply voltage division rule to calculate the initial voltage, $v(0)$.

$$\begin{aligned} v(0) &= \left[\frac{2}{2+1} \right] (18) \\ &= \left[\frac{2}{3} \right] (18) \\ &= 12 \text{ V} \end{aligned}$$

The inductor current is independent of the circuit.

$$i(0) = 0 \text{ A}$$

Step 3 of 10

Draw the modified circuit for $t > 0$.

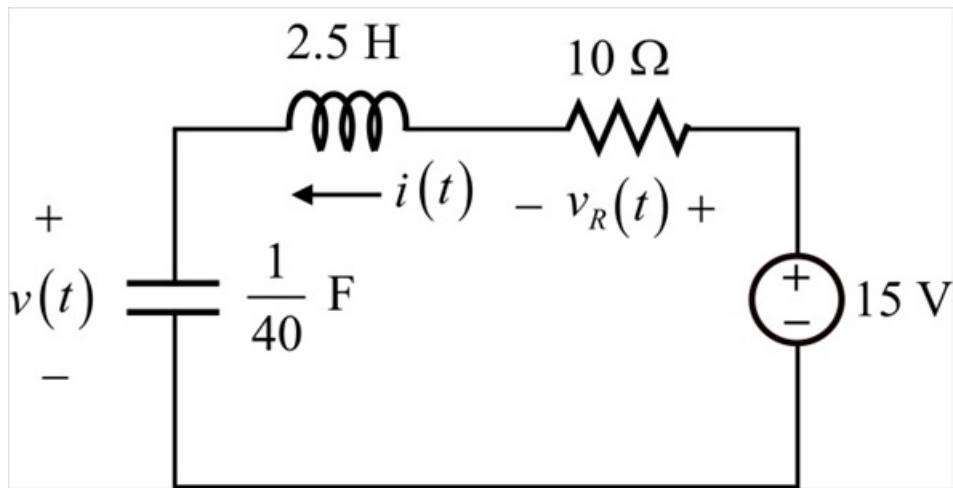


Figure 2

Step 4 of 10

Calculate the value of α .

$$\alpha = \frac{R}{2L}$$

Substitute 10Ω for R and 2.5 H for L .

$$\begin{aligned}\alpha &= \frac{10}{2(2.5)} \\ &= \frac{10}{5} \\ &= 2\end{aligned}$$

Calculate the value of ω_0 .

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute $\frac{1}{40} \text{ F}$ for C and 2.5 H for L .

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{2.5 \cdot \frac{1}{40}}} \\ &= \sqrt{16} \\ &= 4\end{aligned}$$

Step 5 of 10

Since $\omega_0 > \alpha$, the response is under damped.

Calculate the characteristics of the system.

$$\begin{aligned}s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \\ &= -2 \pm \sqrt{4 - 16} \\ &= -2 \pm j3.464\end{aligned}$$

Calculate the total response.

$$v(t) = v_{ss} + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-2t}$$

Here, v_{ss} is the steady state response. It is the final value of the capacitor voltage.

Step 6 of 10

From the circuit, as $t \rightarrow \infty$, circuit reaches steady state and capacitor is replaced with open circuit and inductor with short circuit.

Draw the modified circuit with for $t \rightarrow \infty$.

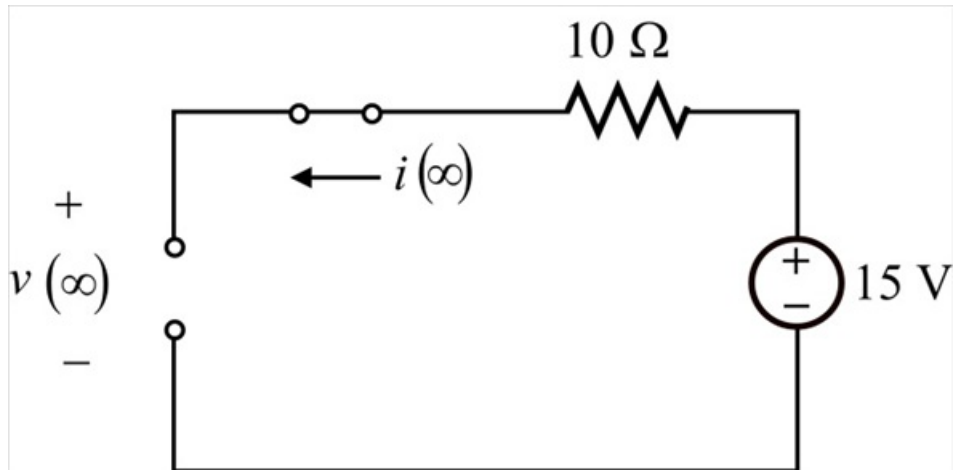


Figure 3

Step 7 of 10

The steady state value is,

$$\begin{aligned} v_{ss} &= v(\infty) \\ &= 15 \text{ V} \end{aligned}$$

Substitute 15 V for v_{ss} in $v(t)$.

$$v(t) = 15 + (A_1 \cos 3.464t + A_2 \sin 3.464t) e^{-2t}$$

Substitute 0 for t .

$$\begin{aligned} v(0) &= 15 + (A_1 \cos 3.464(0) + A_2 \sin 3.464(0)) \\ 12 &= 15 + A_1 \\ A_1 &= -3 \end{aligned}$$

Step 8 of 10

At $t = 0^+$, the inductor and capacitor are in series, so the current passing through it is same.

$$\begin{aligned} i(0) &= C \frac{dv(0)}{dt} \\ \frac{dv(0)}{dt} &= \frac{i(0)}{C} \\ \frac{dv(0)}{dt} &= \frac{0}{\left(\frac{1}{40}\right)} \\ \frac{dv(0)}{dt} &= 0 \text{ V/s} \end{aligned}$$

Differentiate the function $v(t)$ with respect to t .

$$\begin{aligned} \frac{dv(t)}{dt} &= \frac{d}{dt} [15 + (A_1 \cos 3.464t + A_2 \sin 3.464t) e^{-2t}] \\ &= \begin{bmatrix} -2e^{-2t} (A_1 \cos 3.464t + A_2 \sin 3.464t) \\ + e^{-2t} (-3.464 A_1 \sin 3.464t + 3.464 A_2 \cos 3.464t) \end{bmatrix} \end{aligned}$$

Substitute 0 for t .

$$\frac{dv(0)}{dt} = -2A_1 + 3.464A_2$$

Substitute 0 for $\frac{dv(0)}{dt}$ and -3 for A_1 .

$$0 = -2(-3) + 3.464A_2$$

$$3.464A_2 = -6$$

$$A_2 = -1.7321$$

Substitute -1.7321 for A_2 and -3 for A_1 in $v(t)$.

$$\begin{aligned} v(t) &= 15 + e^{-2t} [-3 \cos(3.464t) - 1.7321 \sin(3.464t)] \text{ V} \\ &= 15 - e^{-2t} [3 \cos(3.464t) + 1.7321 \sin(3.464t)] \text{ V} \end{aligned}$$

Hence, the voltage across the capacitor is,

$$\boxed{15 - e^{-2t} [3 \cos(3.464t) + 1.7321 \sin(3.464t)] \text{ V}}$$

Step 10 of 10

Calculate the current $i(t)$.

$$\begin{aligned} i(t) &= C \frac{dv(t)}{dt} \\ &= \left(\frac{1}{40} \right) \frac{d}{dt} [15 - e^{-2t} (3 \cos 3.464t + 1.7321 \sin 3.464t)] \\ &= \frac{1}{40} \left[-(-2)e^{-2t} (3 \cos 3.464t + 1.7321 \sin 3.464t) \right. \\ &\quad \left. - e^{-2t} [1.7321(3.464 \cos 3.464t) + 3(-3.464 \sin 3.464t)] \right] \\ &= \frac{1}{40} e^{-2t} [3.4642 \sin 3.464t + 6 \cos 3.464t - 6 \cos 3.464t + 10.392 \sin 3.464t] \end{aligned}$$

$$i(t) = \frac{1}{40} e^{-2t} [13.8562 \sin 3.464t]$$

Calculate the voltage across the resistor.

$$\begin{aligned} v_R(t) &= 10i(t) \\ &= \frac{10}{40} e^{-2t} [13.8562 \sin 3.464t] \\ &= 3.464e^{-2t} \sin 3.464t \text{ V} \end{aligned}$$

Therefore, the voltage across the resistor is $\boxed{3.464e^{-2t} (\sin 3.464t) \text{ V}}$.